

$\text{int } n = 123$
 $\text{sum} = 3 + 2 + 1$
 $= 6$
 $2 = 12 \% 10$
 $n = 1 = 12 / 10$
 $1 = 1 \% 10$
 $0 = 1 / 10$
 $3 = 123 \% 10$
 $n = 12 = 123 / 10$
 $1 + 2 + 3 = 6$

$\text{while}(\text{condition}) \{$
 $\quad // \text{ block of statement}$
 $\}$

$i \rightarrow 10$

$\text{int } i = 1$
 $\text{while}(i \leq 10) \{$
 $\quad \text{cout} \langle i \rangle$
 $\quad i++$
 $\}$

1
 2
 $\vdots$
 10

$\text{int } n = 123$
 $\text{int } \text{sum} = 0$

$n = 123$

$\text{while}(n \neq 0) \{$
 $\quad \text{int } \text{dgt} = n \% 10;$
 $\quad \text{sum} = \text{sum} + \text{dgt};$
 $\quad n = n / 10;$
 $\}$
 $\text{cout} \langle \text{sum} \rangle$

$0 + 3 = 3$
 $3 + 2 = 5$
 $5 + 1 = 6$

$1 / 10 \rightarrow 0$

}
cout (sum)

$$1/10 \rightarrow 0$$

$$\text{reverse} = \cancel{0} \cancel{3} \cancel{3} \underline{321}$$

$$n = \cancel{0} \cancel{1} \cancel{2} \underline{3} \rightarrow 321$$

$$0 * 10 = 0 + 3$$

$$= \underline{3}$$

$$3 * 10 = 30 + 2$$

$$= 32$$

$$32 * 10 = 320 + 1$$

$$= 321$$

$$n = 123$$

$$\text{reverse} = 0$$

while (n != 0) {

$$\text{reverse} = \text{reverse} * 10$$

$$\text{digit} = n \% 10$$

$$\text{reverse} = \text{reverse} + \text{digit}$$

$$n = n / 10;$$

}

cout(reverse)

$$n = 1234$$